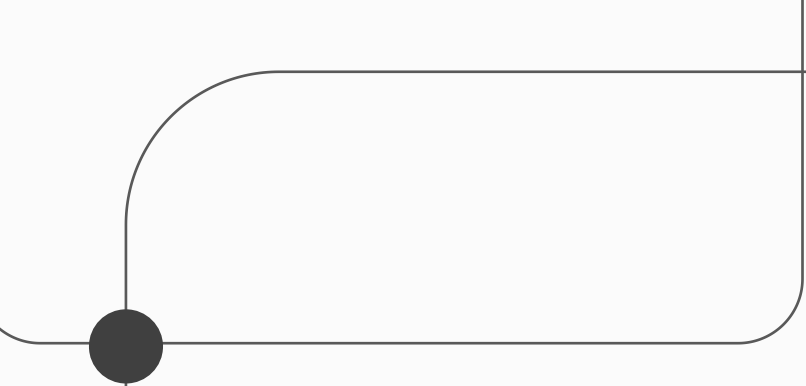




0. Introduction to Quantitative Strategy Design in Cryptocurrency Market (IDS 598.1)

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C O N T E N T S

1. Quantitative Trading

Trending in Quant
Strategies

Think as a Different Role

Decoding Quant Trading

Interdisciplinary
Modeling

Workflow

Mistakes

2. Crypto Market

Blockchain

Crypto Market

Market Making and Uniswap

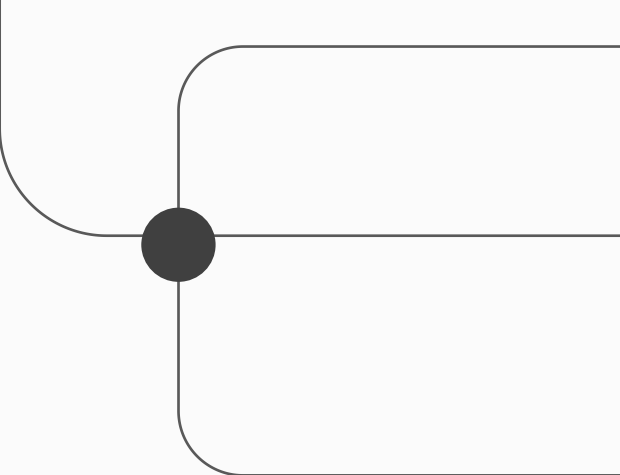
Emerging Crypto Market
(PolyMarket)

Dark side of Cryptos

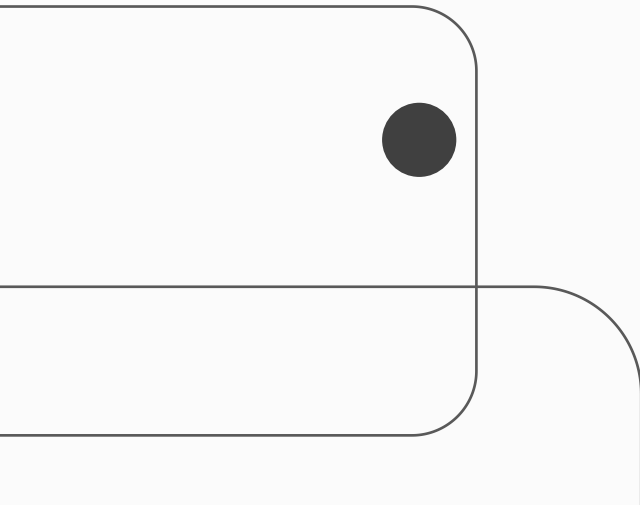
3. Course Policy

Assignment

AI Policy



1. Quantitative Trading



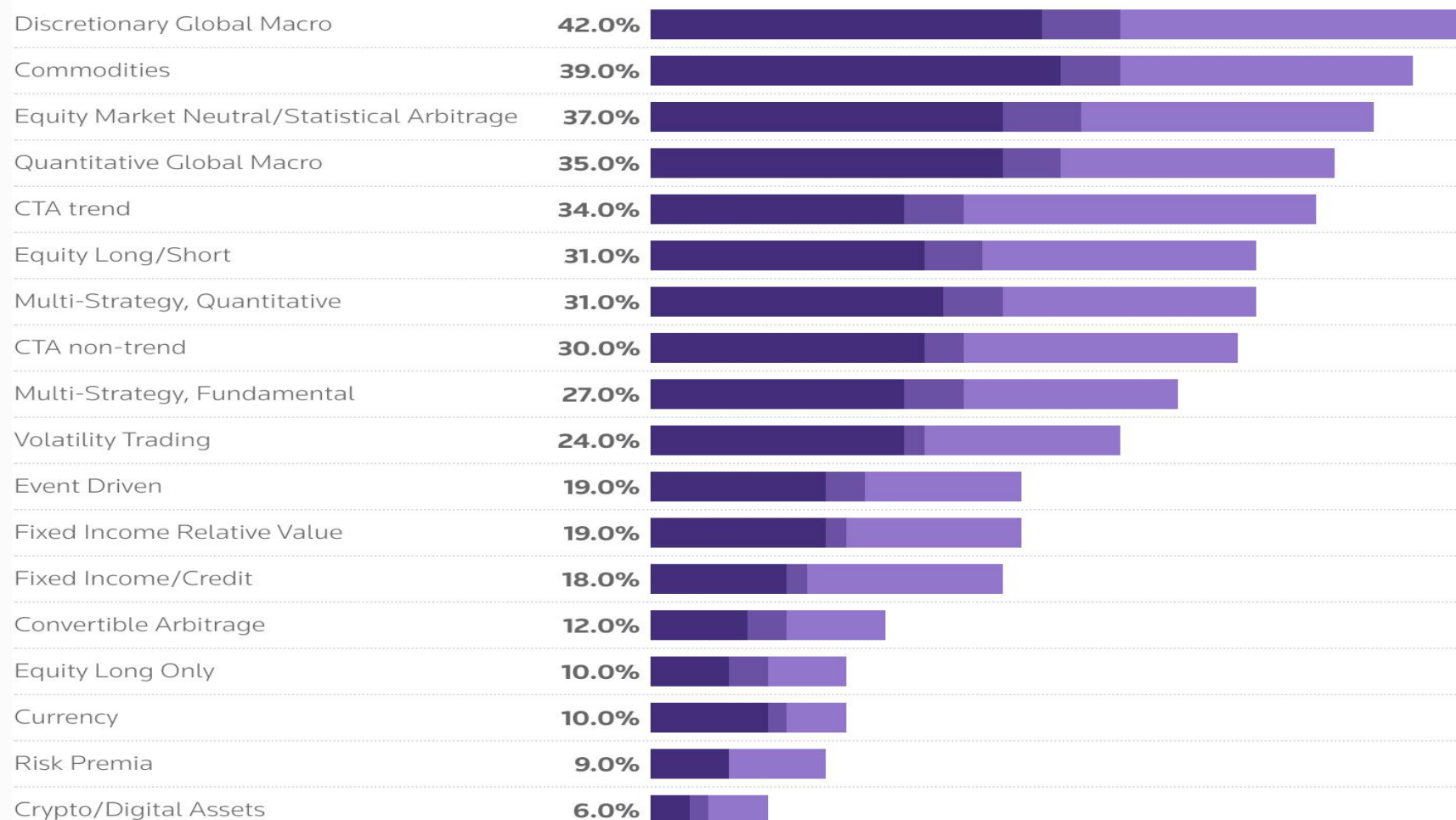
[illegible]

1. Quantitative Trading : Trending in Quant Strategies

Hedge funds wanted

Institutional investors planning allocations within next 12 months, by strategy

● North America ● Asia Pacific ● EMEA

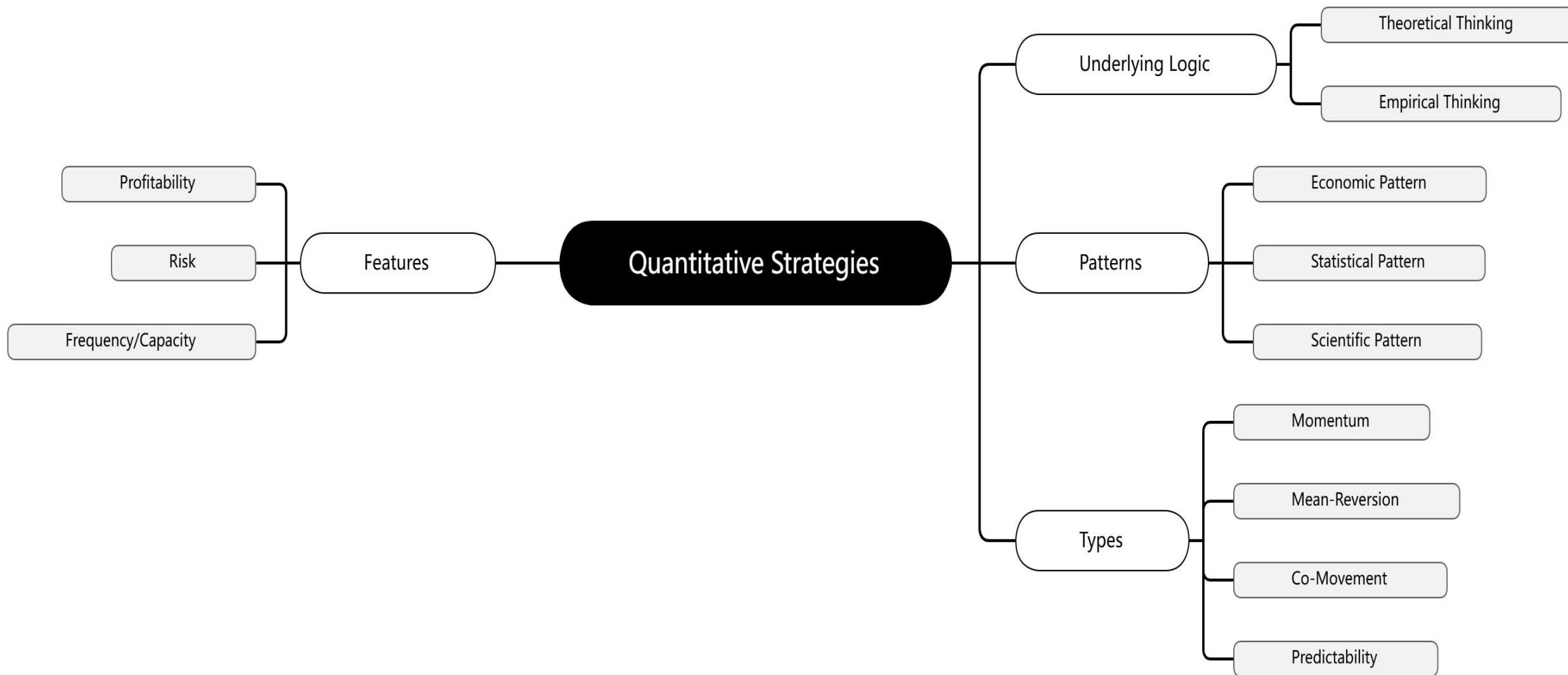


Source: Societe Generale Prime Services & Clearing, as of November 13, 2024 | Chart by Nell Mackenzie

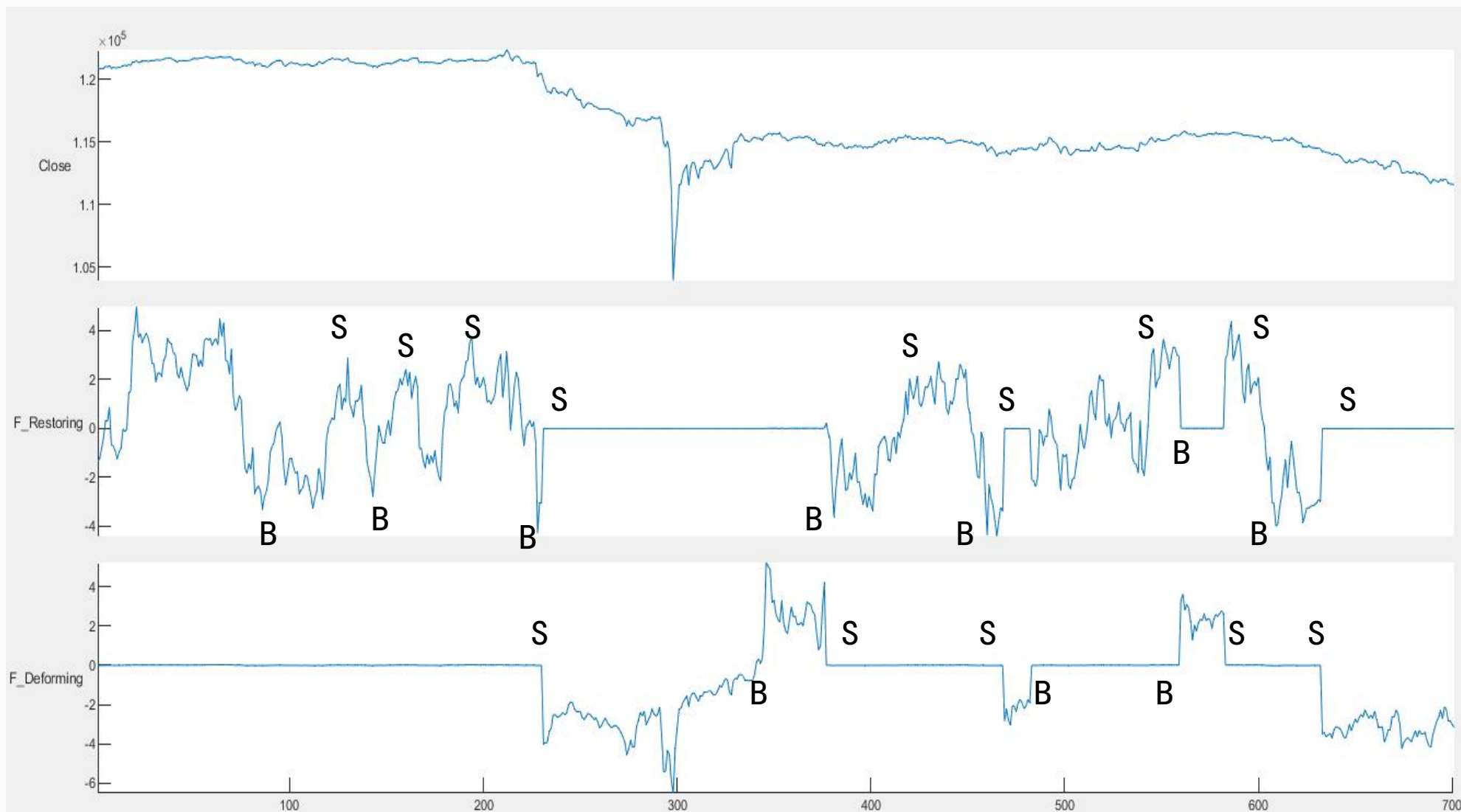
1. Quantitative Trading : Think as a Different Role

	Quantitative Trader	Entrepreneur/Professional/Scientist	Gambler/Speculator
Perception of Risk	Risk is a quantifiable and manageable probability problem.	Risk is the cost of innovation and opportunity; "I am the master of my life."	Praying to gods and relying on luck.
Decision-Making Basis	• Cold-Blooded • Data models • Probabilistic thinking • Strict stop-loss rules	• Enthusiasm • Rational Validation • Resource integration capability • Long-term trend judgment	• Intuition/emotional impulses • "Hot-hand fallacy" (superstition of winning streaks) • Irrational wishful thinking
Primary Motivation	Capital appreciation	Value creation: solving real-world problems + building sustainable contributions/inventions/business	Instant gratification (dopamine rush + short-term excitement from winning)
Attribution of Failure	• Strategy failure • Risk control errors • Market black swan events	• Product-market misfit • Limited time/fundings • Inappropriate approach/management	• "Bad luck" • Perceived dealer manipulation
Time Perspective	Short-to-medium (from nano-seconds to months)	Long-term (from years to decades)	Immediate (from single bets to hours)
Source of Psychological Resilience	Discipline (adhering to trading plans)	Sense of mission	Addiction (inability to stop betting)

1. Quantitative Trading : Decoding Quant Strategies (My own experience)

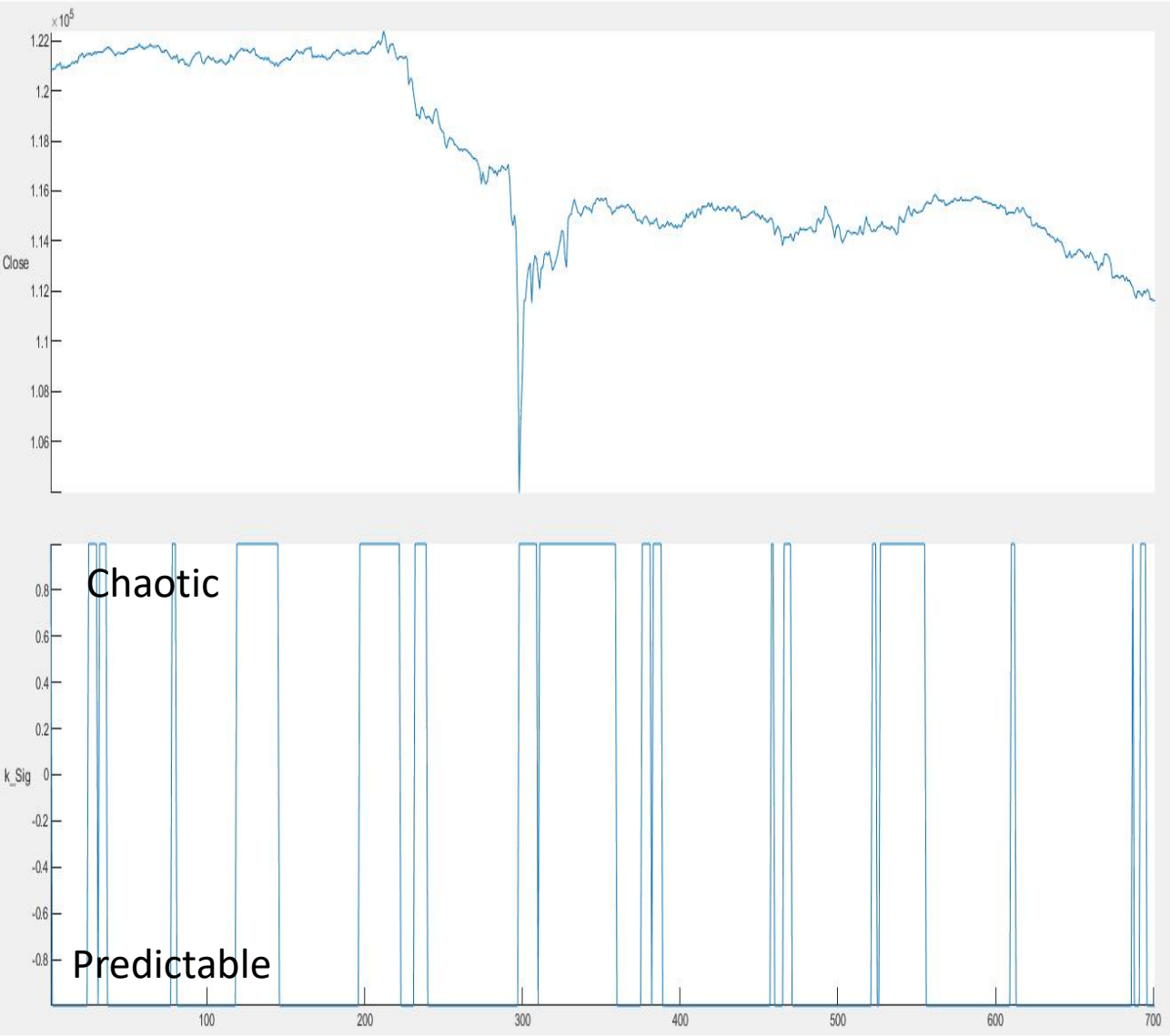


1. Quantitative Trading : Interdisciplinary Modeling: Spring-force model

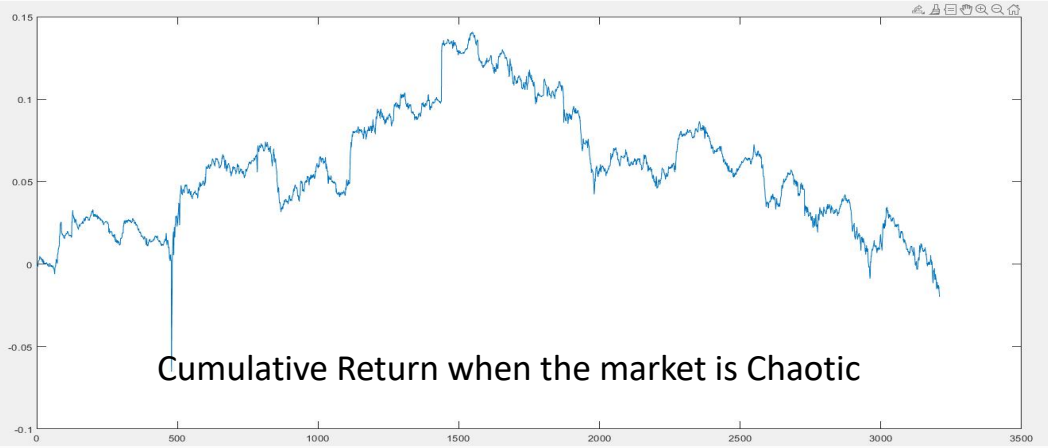


- The signals of mean-reversion and momentum can be separated with segmented springs.

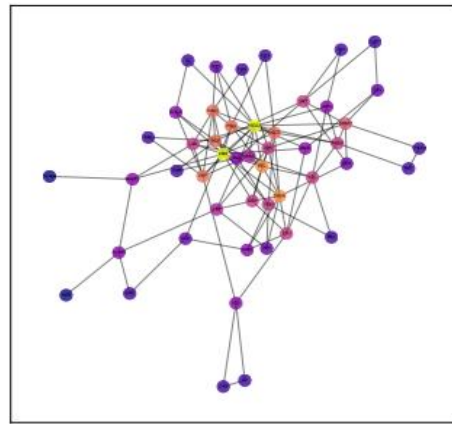
1. Quantitative Trading : Interdisciplinary Modeling: 0-1 Chaotic Test



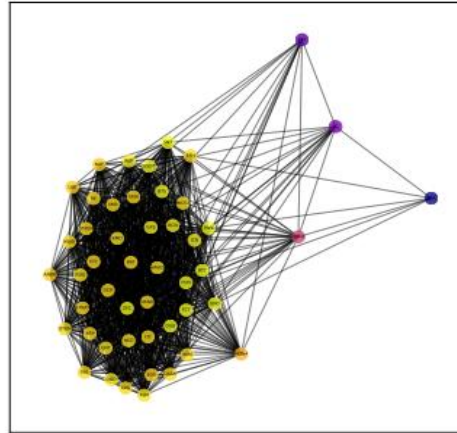
Descriptive Statistics	Mean	Standard Deviation	Skewness	Kurtosis	Cumulative Return
Chaotic	-0.0006%	0.0023	-5.8577	260.2455	-1.99%
Non-chaotic	-0.0043%	0.0016	-0.4747	24.5646	-29.22%



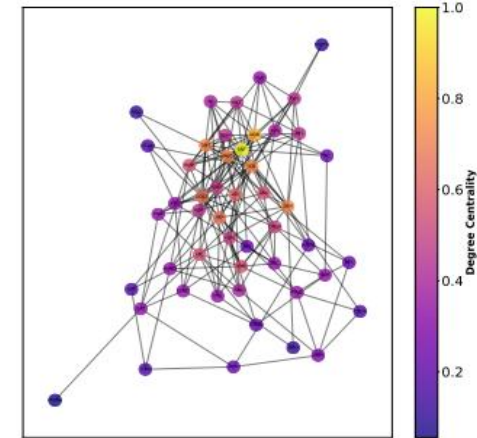
1. Quantitative Trading : Interdisciplinary Modeling: Complex Networks



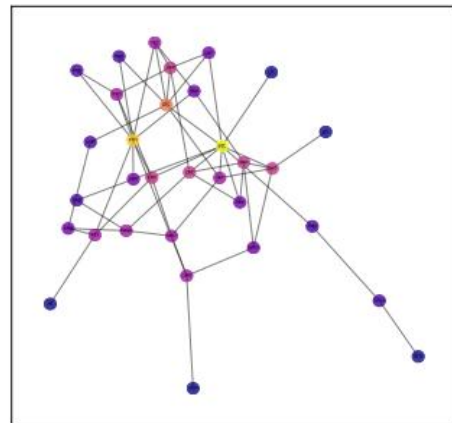
(a) Pre-crash (2017-18)



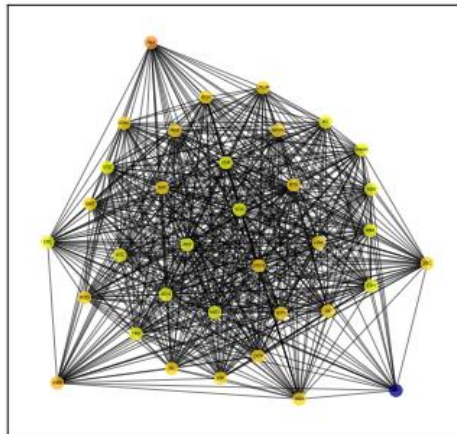
(b) Crash (2017-18)



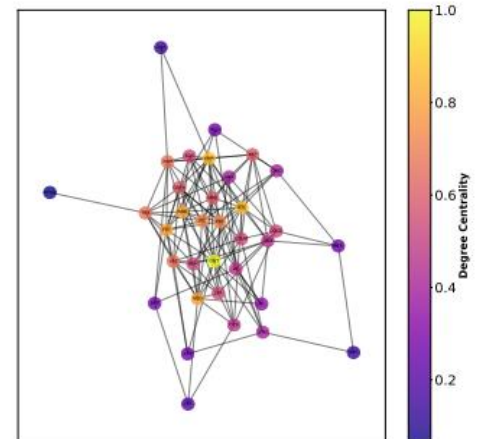
(c) Post-crash (2017-18)



(a) Pre-crash (2018-19)



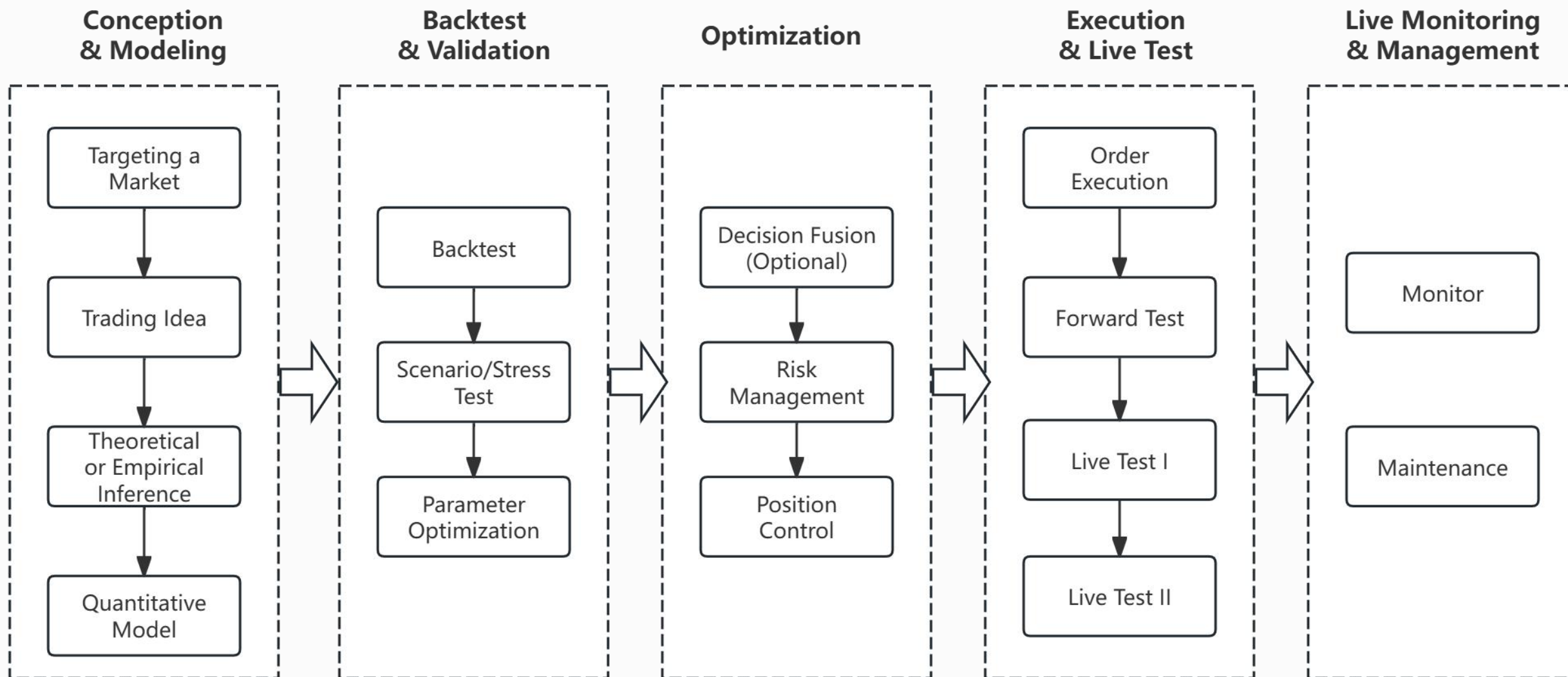
(b) Crash (2018-19)



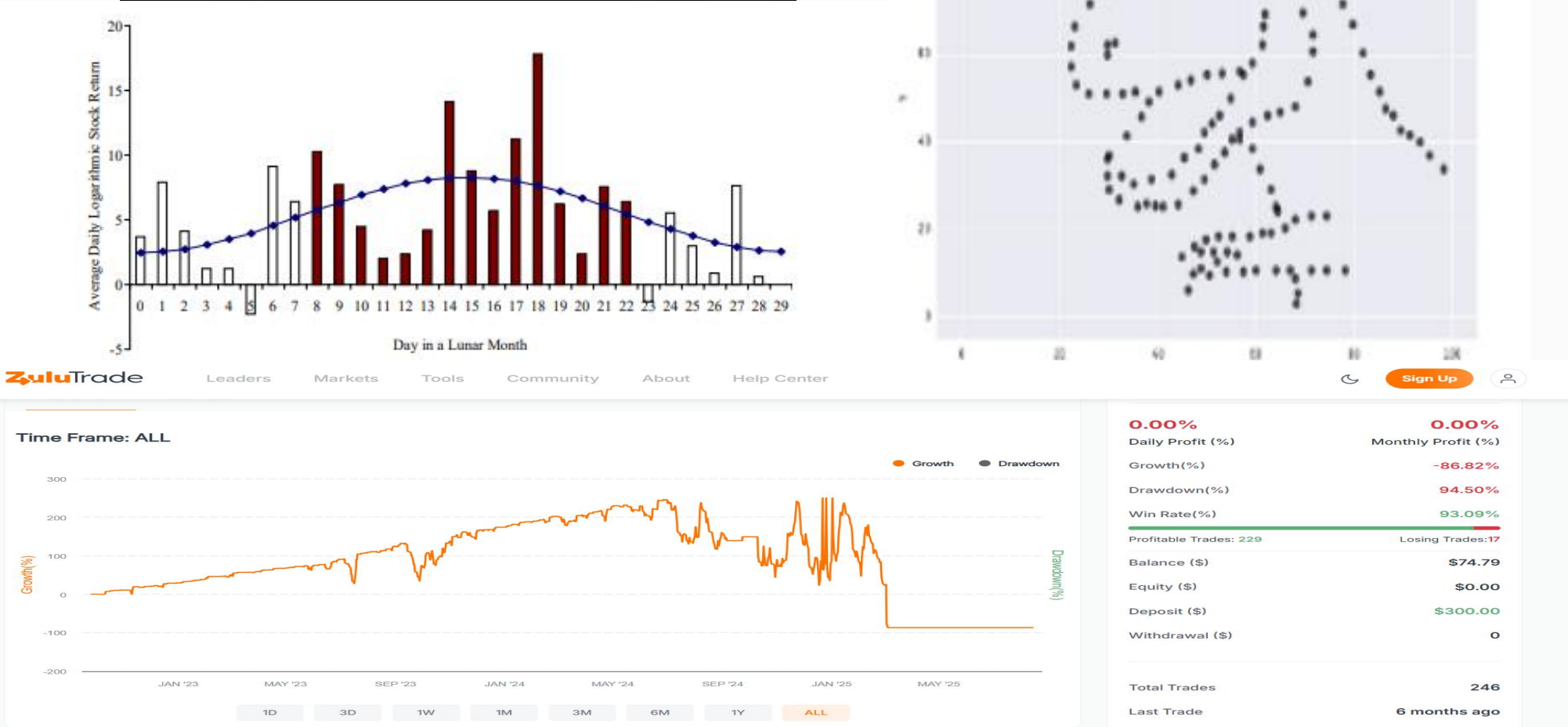
(c) Post-crash (2018-19)

- Mukhia, K., Rai, A., Luwang, S. R., Nurujjaman, M., Majhi, S., & Hens, C. (2024). Complex network analysis of cryptocurrency market during crashes. *Physica A: Statistical Mechanics and its Applications*, 653, 130095.

1. Quantitative Trading : Workflow



1. Quantitative Trading : Mistakes



2. Crypto Market

2. Crypto Market: Blockchain

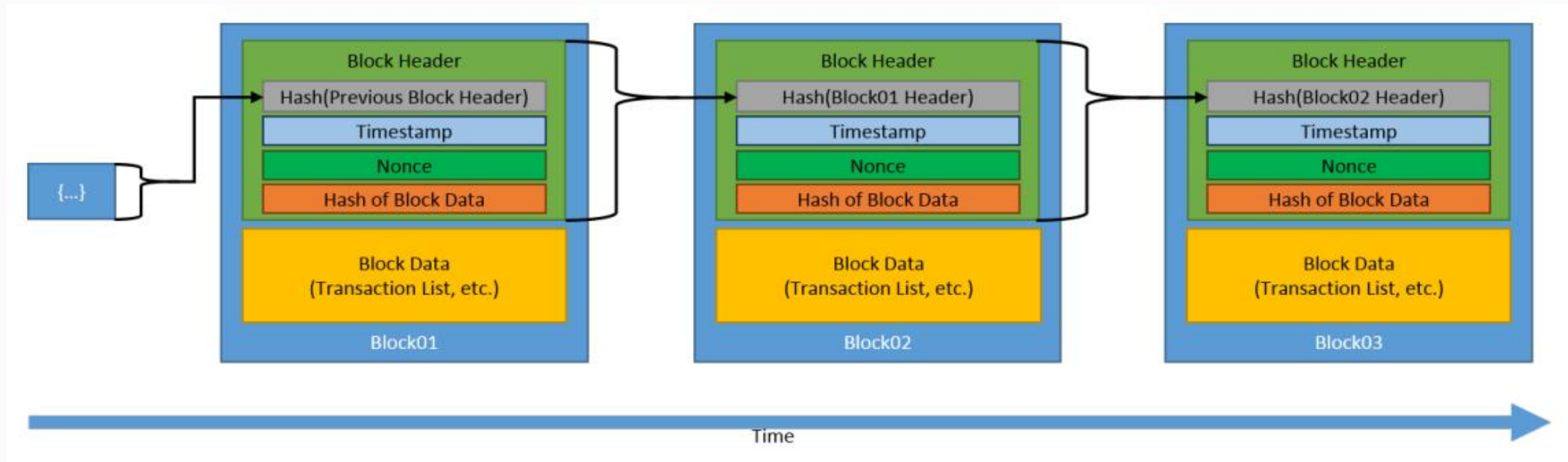
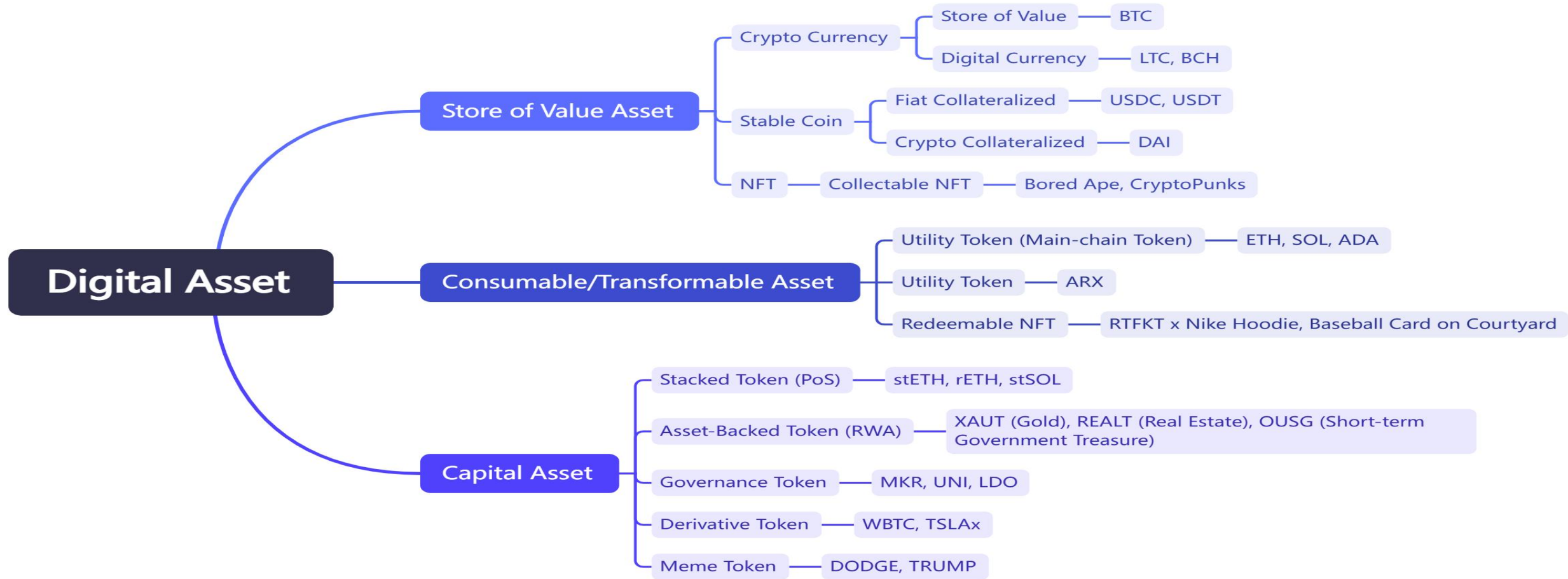
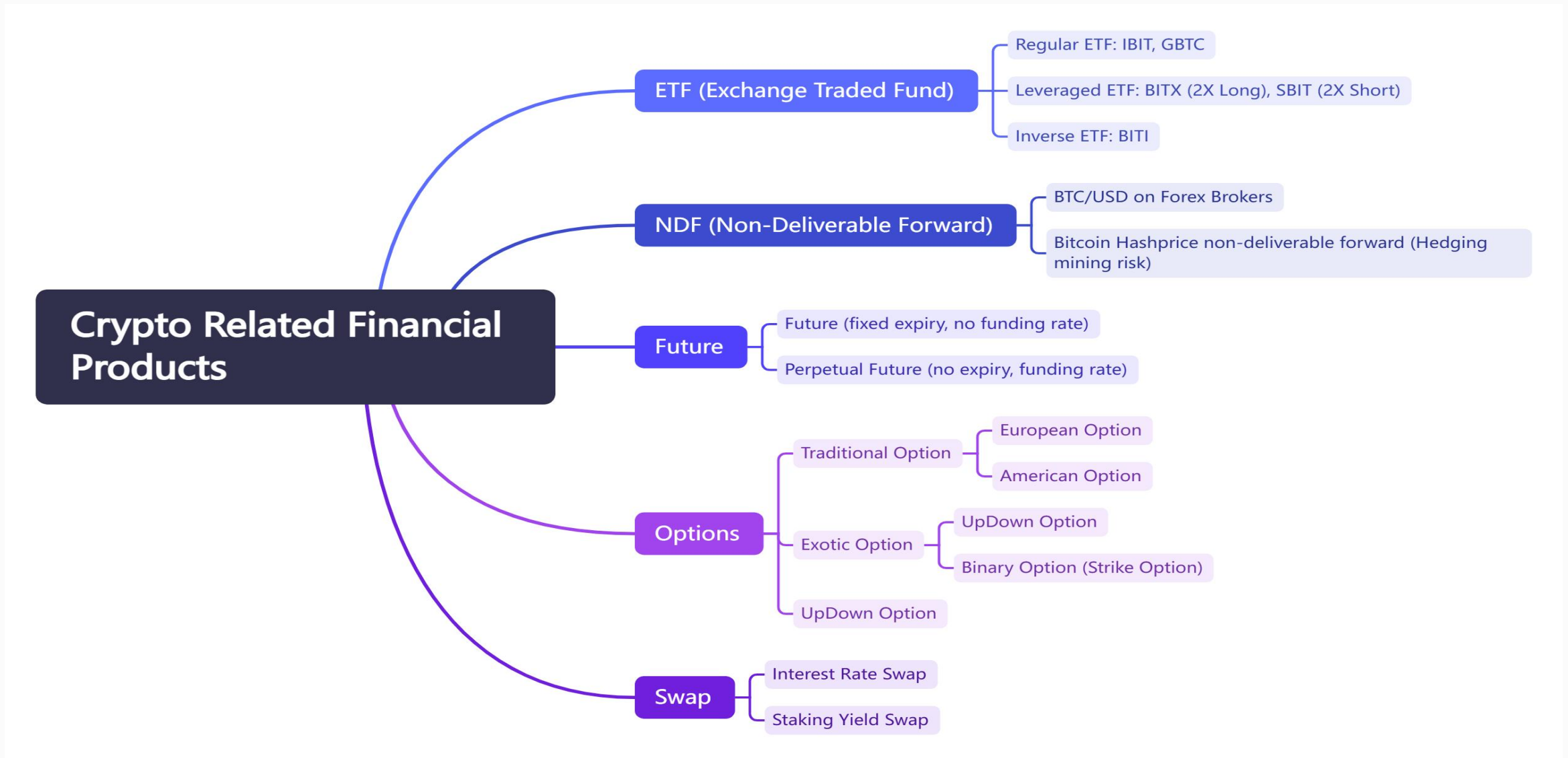


Figure.Generic Chain of Blocks (Yaga et al. 2019)

2. Crypto Market: Digital Assets

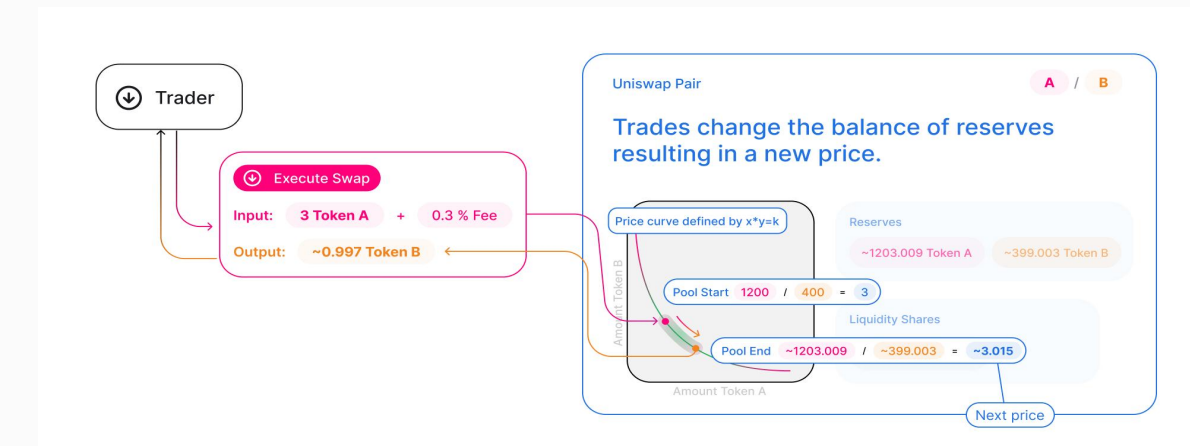
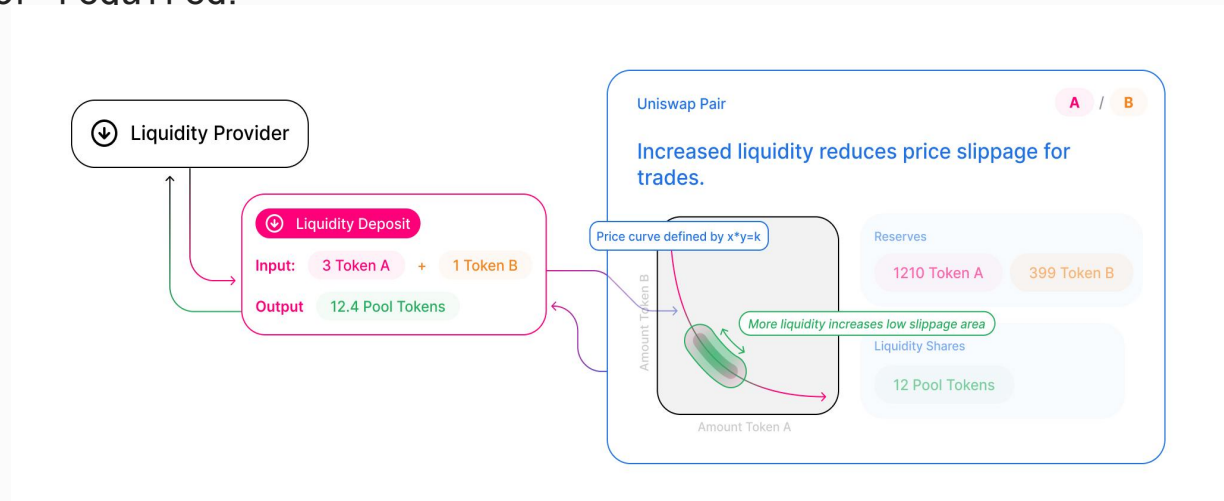
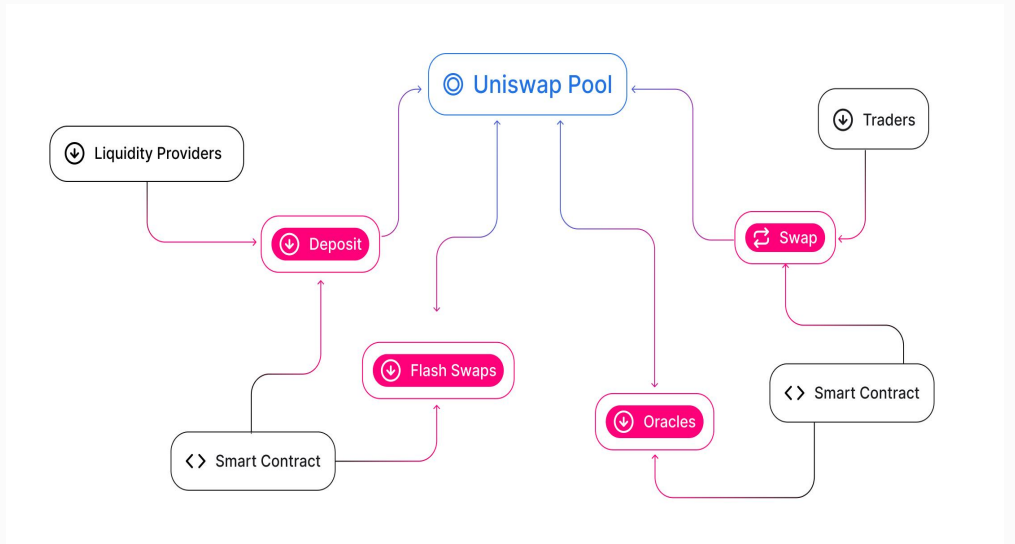


2. Crypto Market: Crypto Related Financial Products

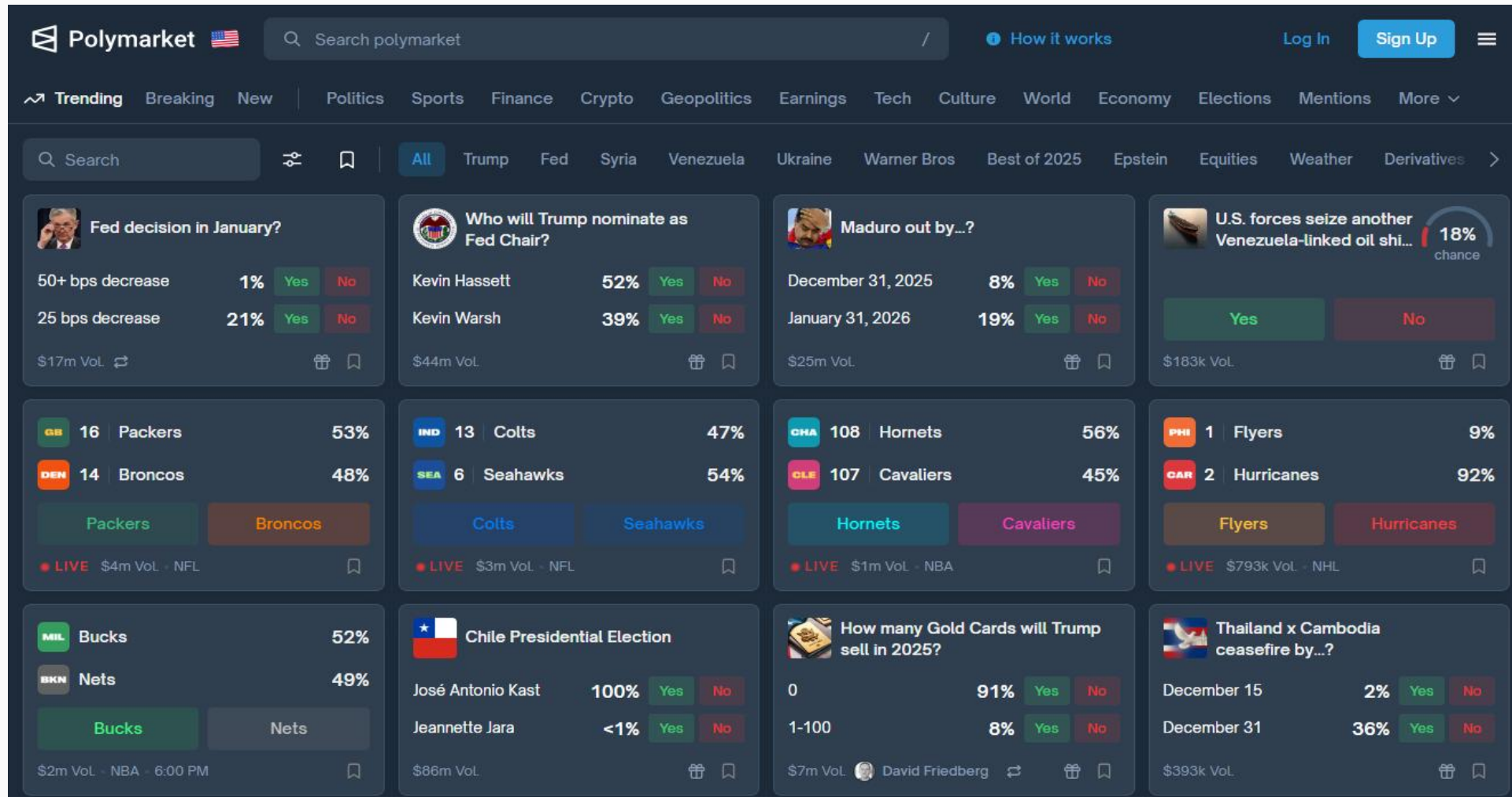


2. Crypto Market: Uniswap and Automated Market Making

At the core of every AMM is a set of smart contracts. These are self-executing programs deployed to public blockchains like Ethereum, and can power trading logic with no centralized operator required.



2. Crypto Market: Prediction Market, Polymarket

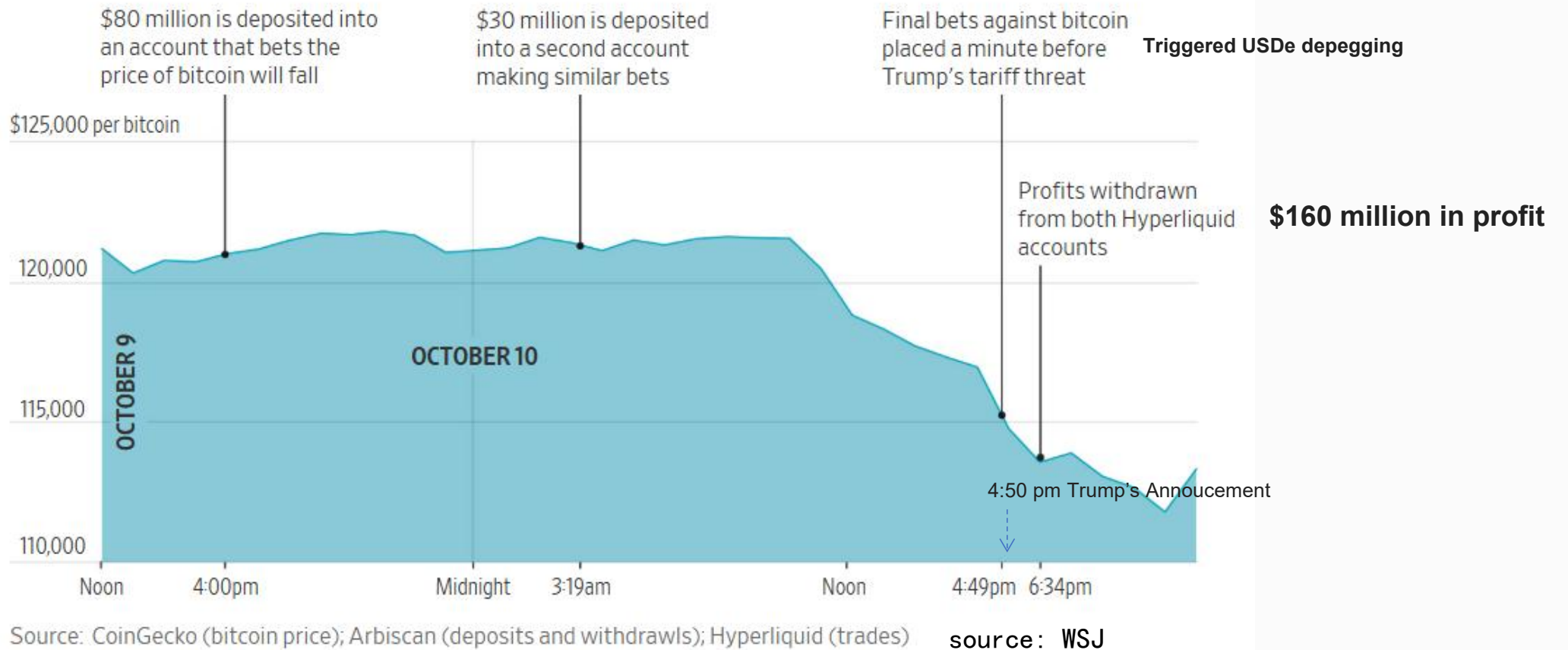


- “Prediction market” is where you can bet on real world events.
- Polymarket is a legally compliant platform regulated by the CFTC, with approval granted in November 2025.

2. Crypto Market: Darkside of Cryptos “Insider Trading?”

Betting bitcoin will fall

Recent trading activity by two Hyperliquid accounts show at least one trader made millions betting the price of bitcoin would decline.



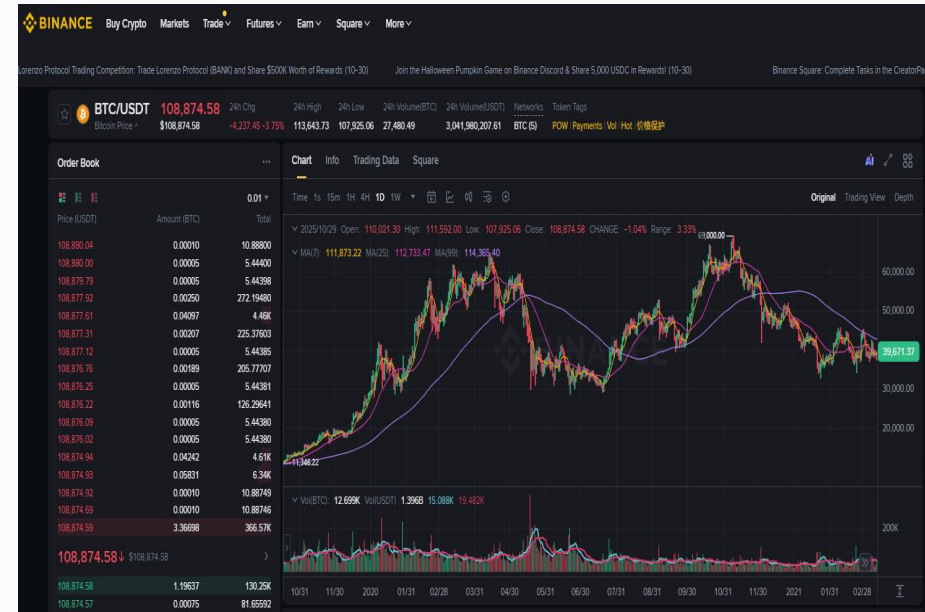
It might be the account deposited \$80 million:

<https://hypurrsan.io/address/0xb317d2bc2d3d2df5fa441b5bae0ab9d8b07283ae>

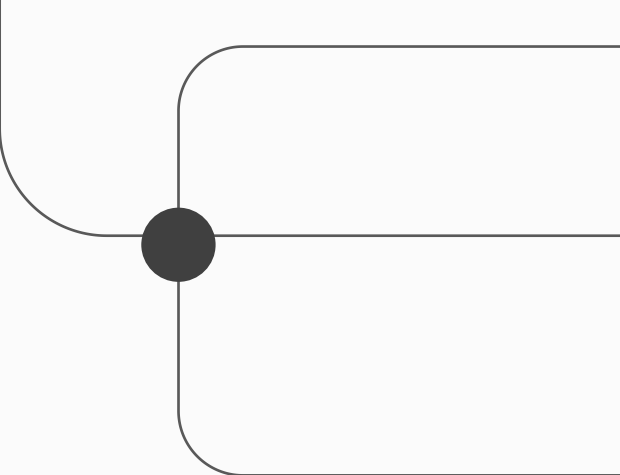
2. Crypto Market: Darkside of Cryptos “Knock Out”



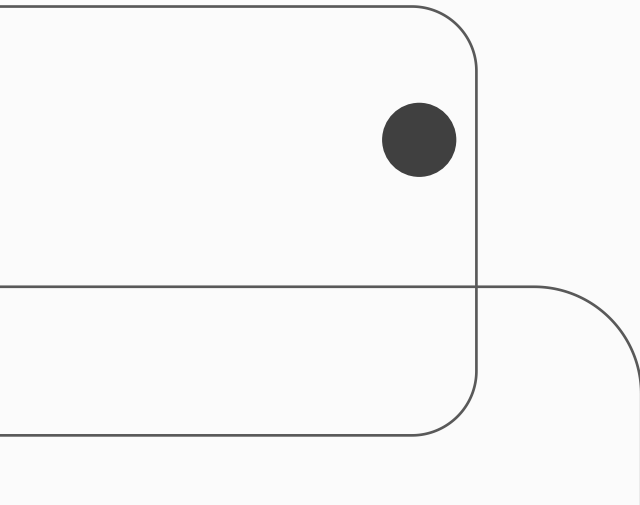
Binance BTC/USDT
(Screenshot taken 2021-10)



Binance BTC/USDT
(Screenshot taken 2025-10-30)



3. Course Policies



3. Course Policies: Assignment, Late and Grading.

- The assignments will due on each Monday.
- It is OK to update your assignments anytime after the due but before the last week of the course. But it is required to keep all the versions of the assignments! I will grade based on the final version.
- Late Policy:
 - Less than 1 day, 10% deduction;
 - 1–5 days, 50% deduction;
 - 5+ days, 100% deduction.

Assignment 1: Crypto Market Analysis	10%
Assignment 2: Trading Idea and Core Logic Design	20%
Assignment 3: Quantitative Model Design	10%
Assignment 4: Empirical Test	10%
Assignment 5: Performance Evaluation and Improvements	10%
? (To be decided) Quiz: Basics of the Crypto Market	20%
Attendance	20%

3. Course Policies: AI Policy

- Originality and novelty in your trading idea are encouraged and will be considered in the grading. However, note that a clear, well-reasoned, and critical analysis of an existing concept can be just as valuable as a completely novel idea.
- It is OK to use AI tools to work with assignments, codes, and build strategies based on AI algorithms.
- **Required Disclosure (Mandatory):**
If you use any generative AI tool (e.g., ChatGPT, Claude, Gemini, Copilot) during any phase of this assignment, including brainstorming, outlining, drafting, or editing.
- **Required Transparency (Mandatory):**
you MUST submit a complete and truthful "Process Appendix" alongside your assignments. **All prompts and responses are required.**
- Prompt Engineering for Trading Strategy Design can be a research project. It is NOT necessary to use it for plagiarism.

The image features a minimalist design on a light gray background. A large, rounded rectangle with a thin gray border is centered. In the top-left corner of this rectangle, there is a solid black circle. In the bottom-right corner, there is another solid black circle. A horizontal black bar is positioned below the text. The text "Thank You" is written in a clean, sans-serif font, with each letter spaced out.

T h a n k Y o u

